🏆 Infosys SE Elite Crack Vault — 06 College ITM Numeric Puzzle Solving 28Q Exam Compendium Elite

🎯 Infosys Systems Engineer (SE) | 📊 19 Questions | 📅 5-6 Nov 2026 Drive | ⚡ 100% Pure SE Verified

Problem 1 • Infosys SE Question 1

MUST-SOLVE

ID: Q1 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q51)
Section: Section 5: Numeric Puzzle Solving | **Page:** 6
Visual Reference Plate: paper_01_page_06.png

Puzzle Statement:
Matchstick. A 2 × 2 grid of squares is made using 12 matchsticks (giving 4 small squares and 1 large square). What is the minimum number of sticks to remove so that no square of any size remains?

Visual Schematic / Puzzle Diagram:

```
+---+---+
|   |   |
+---+---+
|   |   |
+---+---+
12 matchsticks forming four 1×1 squares and one 2×2 square.
```

Visual Schematic / Puzzle Diagram:

```
+---+---+
|   |   |
+---+---+
|   |   |
+---+---+
12 matchsticks forming four 1×1 squares and one 2×2 square.
```

Visual Schematic / Puzzle Diagram:

```
+---+---+
|   |   |
+---+---+
|   |   |
+---+---+
12 matchsticks forming four 1×1 squares and one 2×2 square.
```

Visual Schematic / Puzzle Diagram:

```
+---+---+
|   |   |
+---+---+
|   |   |
+---+---+
12 matchsticks forming four 1×1 squares and one 2×2 square.
```

A 2

B 3

C 4

D 5

✓ **Verified Answer:** (B)** — `3` **Detailed Solution & Formula Deduction:** In a 2 × 2 matchstick grid (12 sticks total, forming four 1×1 squares and one 2×2 square): To eliminate all 5 squares with the minimum number of removals, we need 3 sticks. Removing 2 sticks can at most destroy four 1×1 squares (by removing the two interior central sticks), but the outer perimeter still forms the complete 2×2 square. To break both the four 1×1 squares and the outer 2×2 square, at least 3 sticks must be removed (e.g., one top outer stick, one bottom outer stick, and one internal vertical stick). Minimum = 3.

Problem 2 • Infosys SE Question 2

MUST-SOLVE

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q52)
Section: Section 5: Numeric Puzzle Solving | **Page:** 6
Visual Reference Plate: paper_01_page_06.png

Puzzle Statement:
Circular wheel. Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.
Wheel 1: T = 3, R = 5, B = 4, L = 7, Centre = 14
Wheel 2: T = 2, R = 6, B = 5, L = 3, Centre = 7
Wheel 3: T = 4, R = 3, B = 6, L = 9, Centre = ?

Visual Schematic / Puzzle Diagram:

```
     Wheel 1             Wheel 2             Wheel 3
       [ 3 ]               [ 2 ]               [ 4 ]
         |                   |                   |
[ 7 ]-( 14 )-[ 5 ]   [ 3 ]-(  7  )-[ 6 ]   [ 9 ]-(  ?  )-[ 3 ]
         |                   |                   |
       [ 4 ]               [ 5 ]               [ 6 ]
```

Visual Schematic / Puzzle Diagram:

```
     Wheel 1             Wheel 2             Wheel 3
       [ 3 ]               [ 2 ]               [ 4 ]
         |                   |                   |
[ 7 ]-( 14 )-[ 5 ]   [ 3 ]-(  7  )-[ 6 ]   [ 9 ]-(  ?  )-[ 3 ]
         |                   |                   |
       [ 4 ]               [ 5 ]               [ 6 ]
```

Visual Schematic / Puzzle Diagram:

```
     Wheel 1             Wheel 2             Wheel 3
       [ 3 ]               [ 2 ]               [ 4 ]
         |                   |                   |
[ 7 ]-( 14 )-[ 5 ]   [ 3 ]-(  7  )-[ 6 ]   [ 9 ]-(  ?  )-[ 3 ]
         |                   |                   |
       [ 4 ]               [ 5 ]               [ 6 ]
```

Visual Schematic / Puzzle Diagram:

```
     Wheel 1             Wheel 2             Wheel 3
       [ 3 ]               [ 2 ]               [ 4 ]
         |                   |                   |
[ 7 ]-( 14 )-[ 5 ]   [ 3 ]-(  7  )-[ 6 ]   [ 9 ]-(  ?  )-[ 3 ]
         |                   |                   |
       [ 4 ]               [ 5 ]               [ 6 ]
```

Visual Schematic / Puzzle Diagram:

```
     Wheel 1             Wheel 2             Wheel 3
       [ 3 ]               [ 2 ]               [ 4 ]
         |                   |                   |
[ 7 ]-( 14 )-[ 5 ]   [ 3 ]-(  7  )-[ 6 ]   [ 9 ]-(  ?  )-[ 3 ]
         |                   |                   |
       [ 4 ]               [ 5 ]               [ 6 ]
```

A 21

B 27

C 30

D 33

✓ **Verified Answer:** (C)** — `30` **Detailed Solution & Formula Deduction:** Determining the circular wheel formula: - Wheel 1: T=3, R=5, B=4, L=7, Centre=14. Formula: Centre = (T × B) + L - R = (3 × 4) + 7 - 5 = 12 + 2 = 14 (Checked!) - Wheel 2: T=2, R=6, B=5, L=3, Centre=7. Formula: Centre = (T × B) + L - R = (2 × 5) + 3 - 6 = 10 - 3 = 7 (Checked!) - Wheel 3: T=4, R=3, B=6, L=9, Centre=? Formula: Centre = (T × B) + L - R = (4 × 6) + 9 - 3 = 24 + 6 = 30.

Problem 3 • Infosys SE Question 3

MUST-SOLVE

ID: Q3 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q53)
Section: Section 5: Numeric Puzzle Solving | **Page:** 6
Visual Reference Plate: paper_01_page_06.png

Puzzle Statement:
Matrix logic. Find the missing number.
3
4
25
5
12
169
6
8
?

Visual Schematic / Puzzle Diagram:



Visual Schematic / Puzzle Diagram:



Visual Schematic / Puzzle Diagram:



Visual Schematic / Puzzle Diagram:



```
| 6 | 8 |  ? |
└───┴───┴────┘
```

A 64

B 100

C 196

D 144

✓ **Verified Answer:** (B)** — `100` **Detailed Solution & Formula Deduction:** Matrix logic pattern: Row 1: 3, 4, 25 => $3^2 + 4^2 = 9 + 16 = 25$. Row 2: 5, 12, 169 => $5^2 + 12^2 = 25 + 144 = 169$. Row 3: 6, 8, ? => $6^2 + 8^2 = 36 + 64 = 100$. Missing number = 100.

Problem 4 • Infosys SE Question 4

MUST-SOLVE

ID: Q4 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q54)
Section: Section 5: Numeric Puzzle Solving | **Page:** 7
Visual Reference Plate: paper_01_page_07.png

Puzzle Statement:
Pattern. Find the next term: 2, 12, 36, 80, 150, ?

Visual Schematic / Puzzle Diagram:

```
Terms:        2       12       36       80       150       [ 252 ]
              |        |        |        |         |          |
Formula:   1³+1²    2³+2²    3³+3²    4³+4²    5³+5²      6³+6²
           1 + 1    8 + 4   27 + 9   64 + 16  125 + 25   216 + 36
```

Visual Schematic / Puzzle Diagram:

```
Terms:        2       12       36       80       150       [ 252 ]
              |        |        |        |         |          |
Formula:   1³+1²    2³+2²    3³+3²    4³+4²    5³+5²      6³+6²
           1 + 1    8 + 4   27 + 9   64 + 16  125 + 25   216 + 36
```

Visual Schematic / Puzzle Diagram:

```
Terms:        2       12       36       80       150       [ 252 ]
              |        |        |        |         |          |
Formula:   1³+1²    2³+2²    3³+3²    4³+4²    5³+5²      6³+6²
           1 + 1    8 + 4   27 + 9   64 + 16  125 + 25   216 + 36
```

Visual Schematic / Puzzle Diagram:

```
Terms:        2       12       36       80       150       [ 252 ]
              |        |        |        |         |          |
Formula:   1³+1²    2³+2²    3³+3²    4³+4²    5³+5²      6³+6²
           1 + 1    8 + 4   27 + 9   64 + 16  125 + 25   216 + 36
```

A 216

B 252

C 240

D 294

✓ **Verified Answer:** (B)** — `252` **Detailed Solution & Formula Deduction:** Sequence: 2, 12, 36, 80, 150, ? Pattern: $n^3 + n^2$ for n = 1, 2, 3, 4, 5, ... - n = 1: $1^3 + 1^2 = 1 + 1 = 2$ - n = 2: $2^3 + 2^2 = 8 + 4 = 12$ - n = 3: $3^3 + 3^2 = 27 + 9 = 36$ - n = 4: $4^3 + 4^2 = 64 + 16 = 80$ - n = 5: $5^3 + 5^2 = 125 + 25 = 150$ - n = 6: $6^3 + 6^2 = 216 + 36 = 252$.

Problem 5 • Infosys SE Question 5

MUST-SOLVE

ID: Q5 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q51)
Section: Section 5: Numeric Puzzle Solving | **Page:** 6
Visual Reference Plate: paper_02_page_06.png

Puzzle Statement:
Matchstick. Five identical unit squares are arranged in a row sharing sides, using 16 matchsticks in total. What is the minimum number of sticks that must be removed so that exactly 3 complete unit squares remain?

Visual Schematic / Puzzle Diagram:

```
+---+---+---+---+---+
| 1 | 2 | 3 | 4 | 5 |
+---+---+---+---+---+
5 unit squares in a line built from 16 matchsticks.
```

Visual Schematic / Puzzle Diagram:

```
+---+---+---+---+---+
| 1 | 2 | 3 | 4 | 5 |
+---+---+---+---+---+
5 unit squares in a line built from 16 matchsticks.
```

Visual Schematic / Puzzle Diagram:

```
+---+---+---+---+---+
| 1 | 2 | 3 | 4 | 5 |
+---+---+---+---+---+
5 unit squares in a line built from 16 matchsticks.
```

Visual Schematic / Puzzle Diagram:

```
+---+---+---+---+---+
| 1 | 2 | 3 | 4 | 5 |
+---+---+---+---+---+
5 unit squares in a line built from 16 matchsticks.
```

A 1

B 2

C 3

D 4

✓ **Verified Answer:** (A)** — `1` **Detailed Solution & Formula Deduction:** 1. The stick shared between the 2nd and 3rd squares is a side of both. Removing that single stick breaks both squares at once, leaving the 1st, 4th and 5th squares intact — exactly 3 complete squares, with only 1 stick removed.

Problem 6 • Infosys SE Question 6 MUST-SOLVE

ID: Q6 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q52)
Section: Section 5: Numeric Puzzle Solving | **Page:** 6
Visual Reference Plate: paper_02_page_06.png

Puzzle Statement:
Circular wheel. Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.
Wheel 1: T = 6, R = 2, B = 4, L = 3, Centre = 30
Wheel 2: T = 5, R = 3, B = 8, L = 2, Centre = 46
Wheel 3: T = 7, R = 4, B = 3, L = 6, Centre = ?

Visual Schematic / Puzzle Diagram:

```
     Wheel 1             Wheel 2             Wheel 3
       [ 6 ]               [ 5 ]               [ 7 ]
         |                   |                   |
[ 3 ]-( 30 )-[ 2 ]   [ 2 ]-( 46 )-[ 3 ]   [ 6 ]-(  ?  )-[ 4 ]
         |                   |                   |
       [ 4 ]               [ 8 ]               [ 3 ]
```

Visual Schematic / Puzzle Diagram:

```
     Wheel 1             Wheel 2             Wheel 3
       [ 6 ]               [ 5 ]               [ 7 ]
         |                   |                   |
[ 3 ]-( 30 )-[ 2 ]   [ 2 ]-( 46 )-[ 3 ]   [ 6 ]-(  ?  )-[ 4 ]
         |                   |                   |
       [ 4 ]               [ 8 ]               [ 3 ]
```

Visual Schematic / Puzzle Diagram:

```
     Wheel 1             Wheel 2             Wheel 3
       [ 6 ]               [ 5 ]               [ 7 ]
         |                   |                   |
[ 3 ]-( 30 )-[ 2 ]   [ 2 ]-( 46 )-[ 3 ]   [ 6 ]-(  ?  )-[ 4 ]
         |                   |                   |
       [ 4 ]               [ 8 ]               [ 3 ]
```

Visual Schematic / Puzzle Diagram:

```
     Wheel 1             Wheel 2             Wheel 3
       [ 6 ]               [ 5 ]               [ 7 ]
         |                   |                   |
[ 3 ]-( 30 )-[ 2 ]   [ 2 ]-( 46 )-[ 3 ]   [ 6 ]-(  ?  )-[ 4 ]
         |                   |                   |
       [ 4 ]               [ 8 ]               [ 3 ]
```

A 39

B 41

C 45

D 49

✓ **Verified Answer:** (C)** — `45` **Detailed Solution & Formula Deduction:** 45. Rule: Centre = (T × B) + (R × L). Wheel 1: (6×4)+(2×3) = 24+6 = 30 . Wheel 2: (5×8)+(3×2) = 40+6 = ✓ 46 . Wheel 3: (7×3)+(4×6) = 21+24 = ✓ 45.

Problem 7 • Infosys SE Question 7

MUST-SOLVE

ID: Q7 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q53)
Section: Section 5: Numeric Puzzle Solving | **Page:** 6
Visual Reference Plate: paper_02_page_06.png

Puzzle Statement:
Matrix logic. Find the missing number.
2
3
11
3
4
19
4
5
?

Visual Schematic / Puzzle Diagram:



Visual Schematic / Puzzle Diagram:



Visual Schematic / Puzzle Diagram:



Visual Schematic / Puzzle Diagram:



```
| 4 | 5 |  ? |
└───┴───┴────┘
```

A 27

B 29

C 31

D 33

✓ **Verified Answer:** (B)** — `29` **Detailed Solution & Formula Deduction:** 29. Rule: third column = (a×b) + a + b. Row 1: (2×3)+2+3 = 11 . Row 2: (3×4)+3+4 = 19 . Row 3: ✓ ✓ (4×5)+4+5 = 20+9 = 29.

Problem 8 • Infosys SE Question 8

CORE SE

ID: Q8 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q54)
Section: Section 5: Numeric Puzzle Solving | **Page:** 6
Visual Reference Plate: paper_02_page_06.png

Puzzle Statement:
Pattern. Find the next term: 1, 4, 10, 20, 35, ?

Visual Schematic / Puzzle Diagram:

```
Terms:        1        4       10       20       35       [ 56 ]
1st Diff:         +3       +6       +10      +15      [ +21 ]
2nd Diff:             +3       +4       +5       [ +6 ]
```

Visual Schematic / Puzzle Diagram:

```
Terms:        1        4       10       20       35       [ 56 ]
1st Diff:         +3       +6       +10      +15      [ +21 ]
2nd Diff:             +3       +4       +5       [ +6 ]
```

Visual Schematic / Puzzle Diagram:

```
Terms:        1        4       10       20       35       [ 56 ]
1st Diff:         +3       +6       +10      +15      [ +21 ]
2nd Diff:             +3       +4       +5       [ +6 ]
```

Visual Schematic / Puzzle Diagram:

```
Terms:        1        4       10       20       35       [ 56 ]
1st Diff:         +3       +6       +10      +15      [ +21 ]
2nd Diff:             +3       +4       +5       [ +6 ]
```

A 52

B 56

C 60

D 49

✓ **Verified Answer:** (B)** — `56` **Detailed Solution & Formula Deduction:** 56. These are tetrahedral numbers, C(n+2,3): 1, 4, 10, 20, 35, and next is C(8,3) = 56. Section 6: Grammar

Problem 9 • Infosys SE Question 9

CORE SE

ID: Q9 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q51)
Section: Section 5: Numeric Puzzle Solving | **Page:** 7
Visual Reference Plate: paper_03_page_07.png

Puzzle Statement:
Matchstick. A figure is made of 3 unit squares arranged in an L-shape (one square with a second attached to its right, and a third attached below the first), using 10 matchsticks in total. What is the minimum number of sticks to remove so that no complete square remains?

Visual Schematic / Puzzle Diagram:

```
+---+---+
| 1 | 2 |
+---+---+
| 3 |
+---+
L-shaped figure of 3 squares made of 10 matchsticks.
```

Visual Schematic / Puzzle Diagram:

```
+---+---+
| 1 | 2 |
+---+---+
| 3 |
+---+
L-shaped figure of 3 squares made of 10 matchsticks.
```

Visual Schematic / Puzzle Diagram:

```
+---+---+
| 1 | 2 |
+---+---+
| 3 |
+---+
L-shaped figure of 3 squares made of 10 matchsticks.
```

Visual Schematic / Puzzle Diagram:

```
+---+---+
| 1 | 2 |
+---+---+
| 3 |
+---+
L-shaped figure of 3 squares made of 10 matchsticks.
```

A 1

B 2

C 3

D 4

✓ **Verified Answer:** (B)** — `2` **Detailed Solution & Formula Deduction:** 2. The stick shared between the top-left square and the top-right square is also a side of both. Removing it breaks both squares in one move. One more removal (any exclusive side of the bottom square) breaks the third — 2 sticks minimum.

Problem 10 • Infosys SE Question 10

CORE SE

ID: Q10 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q52)
Section: Section 5: Numeric Puzzle Solving | **Page:** 7
Visual Reference Plate: paper_03_page_07.png

Puzzle Statement:
Circular wheel. Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.
Wheel 1: T = 5, R = 3, B = 7, L = 2, Centre = 19
Wheel 2: T = 4, R = 6, B = 9, L = 3, Centre = 63
Wheel 3: T = 8, R = 5, B = 6, L = 4, Centre = ?

Visual Schematic / Puzzle Diagram:

```
     Wheel 1             Wheel 2             Wheel 3
       [ 5 ]               [ 4 ]               [ 8 ]
         |                   |                   |
[ 2 ]-( 19 )-[ 3 ]   [ 3 ]-( 63 )-[ 6 ]   [ 4 ]-(  ?  )-[ 5 ]
         |                   |                   |
       [ 7 ]               [ 9 ]               [ 6 ]
```

Visual Schematic / Puzzle Diagram:

```
     Wheel 1             Wheel 2             Wheel 3
       [ 5 ]               [ 4 ]               [ 8 ]
         |                   |                   |
[ 2 ]-( 19 )-[ 3 ]   [ 3 ]-( 63 )-[ 6 ]   [ 4 ]-(  ?  )-[ 5 ]
         |                   |                   |
       [ 7 ]               [ 9 ]               [ 6 ]
```

Visual Schematic / Puzzle Diagram:

```
     Wheel 1             Wheel 2             Wheel 3
       [ 5 ]               [ 4 ]               [ 8 ]
         |                   |                   |
[ 2 ]-( 19 )-[ 3 ]   [ 3 ]-( 63 )-[ 6 ]   [ 4 ]-(  ?  )-[ 5 ]
         |                   |                   |
       [ 7 ]               [ 9 ]               [ 6 ]
```

Visual Schematic / Puzzle Diagram:

```
     Wheel 1             Wheel 2             Wheel 3
       [ 5 ]               [ 4 ]               [ 8 ]
         |                   |                   |
[ 2 ]-( 19 )-[ 3 ]   [ 3 ]-( 63 )-[ 6 ]   [ 4 ]-(  ?  )-[ 5 ]
         |                   |                   |
       [ 7 ]               [ 9 ]               [ 6 ]
```

A 54

B 58

C 62

D 50

✓ **Verified Answer:** (B)** — `58` **Detailed Solution & Formula Deduction:** 58. Rule: Centre = (T+B)×(R+L) − (T×B + R×L). Wheel 1: (5+7)(3+2)−(5×7+3×2) = 60−41 = 19 ✓. Wheel 2: (4+9)(6+3)−(4×9+6×3) = 117−54 = 63 ✓. Wheel 3: (8+6)(5+4)−(8×6+5×4) = 126−68 = 58.

Problem 11 • Infosys SE Question 11

CORE SE

ID: Q11 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q53)
Section: Section 5: Numeric Puzzle Solving | **Page:** 7
Visual Reference Plate: paper_03_page_07.png

Puzzle Statement:
Matrix logic. Find the missing number.

4
9
25
9
25
49
25
49
?

Visual Schematic / Puzzle Diagram:

4	9	25
9	25	49
25	49	?

Visual Schematic / Puzzle Diagram:

4	9	25
9	25	49
25	49	?

Visual Schematic / Puzzle Diagram:

4	9	25
9	25	49
25	49	?

Visual Schematic / Puzzle Diagram:

4	9	25
9	25	49

```
| 25 | 49 |  ?  |
└────┴────┴─────┘
```

A 100

B 121

C 144

D 81

✓ **Verified Answer:** (B)** — `121` **Detailed Solution & Formula Deduction:** 121. Each entry is the square of a prime, arranged so that reading along each anti-diagonal gives consecutive primes: $2^2, 3^2, 5^2, 7^2, 11^2$. The missing cell is $11^2 = 121$.

Problem 12 • Infosys SE Question 12

CORE SE

ID: Q12 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q54)
Section: Section 5: Numeric Puzzle Solving | **Page:** 7
Visual Reference Plate: paper_03_page_07.png

Puzzle Statement:
Pattern. Find the next term: 6, 24, 60, 120, 210, ?

Visual Schematic / Puzzle Diagram:

```
Terms:        6       24       60       120       210       [ 336 ]
              |        |        |         |         |          |
Formula:   1·2·3    2·3·4    3·4·5     4·5·6     5·6·7      6·7·8
           1³ - 1   2³ - 2   3³ - 3   4³ - 4   5³ - 5      6³ - 6
```

Visual Schematic / Puzzle Diagram:

```
Terms:        6       24       60       120       210       [ 336 ]
              |        |        |         |         |          |
Formula:   1·2·3    2·3·4    3·4·5     4·5·6     5·6·7      6·7·8
           1³ - 1   2³ - 2   3³ - 3   4³ - 4   5³ - 5      6³ - 6
```

Visual Schematic / Puzzle Diagram:

```
Terms:        6       24       60       120       210       [ 336 ]
              |        |        |         |         |          |
Formula:   1·2·3    2·3·4    3·4·5     4·5·6     5·6·7      6·7·8
           1³ - 1   2³ - 2   3³ - 3   4³ - 4   5³ - 5      6³ - 6
```

Visual Schematic / Puzzle Diagram:

```
Terms:        6       24       60       120       210       [ 336 ]
              |        |        |         |         |          |
Formula:   1·2·3    2·3·4    3·4·5     4·5·6     5·6·7      6·7·8
           1³ - 1   2³ - 2   3³ - 3   4³ - 4   5³ - 5      6³ - 6
```

A 320

B 330

C 336

D 342

✓ **Verified Answer:** (C)** — `336` **Detailed Solution & Formula Deduction:** 336. Each term is the product of 3 consecutive integers: 1×2×3, 2×3×4, 3×4×5, 4×5×6, 5×6×7; next is 6×7×8 = 336. Section 6: Grammar

Problem 13 • Infosys SE Question 13

CORE SE

ID: Q13 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q51)
Section: Section 5: Numeric Puzzle Solving | **Page:** 7
Visual Reference Plate: paper_04_page_07.png

Puzzle Statement:
Matchstick. A 3×3 grid of unit squares (9 small squares) is built using 24 matchsticks. Counting squares of every size (1×1, 2×2 and 3×3 — 14 squares in total), what is the minimum number of matchsticks that must be removed so that no complete square of any size remains?

Visual Schematic / Puzzle Diagram:

```
+---+---+---+
| 1 | 2 | 3 |
+---+---+---+
| 4 | 5 | 6 |
+---+---+---+
| 7 | 8 | 9 |
+---+---+---+
3×3 grid of 9 small squares made of 24 matchsticks.
```

Visual Schematic / Puzzle Diagram:

```
+---+---+---+
| 1 | 2 | 3 |
+---+---+---+
| 4 | 5 | 6 |
+---+---+---+
| 7 | 8 | 9 |
+---+---+---+
3×3 grid of 9 small squares made of 24 matchsticks.
```

Visual Schematic / Puzzle Diagram:

```
+---+---+---+
| 1 | 2 | 3 |
+---+---+---+
| 4 | 5 | 6 |
+---+---+---+
| 7 | 8 | 9 |
+---+---+---+
3×3 grid of 9 small squares made of 24 matchsticks.
```

Visual Schematic / Puzzle Diagram:

```
+---+---+---+
| 1 | 2 | 3 |
+---+---+---+
| 4 | 5 | 6 |
+---+---+---+
| 7 | 8 | 9 |
+---+---+---+
3×3 grid of 9 small squares made of 24 matchsticks.
```

A 4

B 5

C 6

D 7

✓ **Verified Answer:** (C)** — `6` **Detailed Solution & Formula Deduction:** 6. This was verified by exhaustive computer search over all 24 sticks and all 14 squares (nine 1×1, four 2×2, one 3×3): no set of 5 or fewer sticks intersects every square's stick-set, but a 6-stick set exists that does (for example, removing the sticks at positions H(0,0), H(1,0), H(1,1), H(2,1), H(0,2) and V(2,2) breaks all 14 squares). Minimum = 6.

Problem 14 • Infosys SE Question 14

CORE SE

ID: Q14 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q52)
Section: Section 5: Numeric Puzzle Solving | **Page:** 7
Visual Reference Plate: paper_04_page_07.png

Puzzle Statement:
Circular wheel. Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.
Wheel 1: T = 4, R = 5, B = 6, L = 3, Centre = 39
Wheel 2: T = 7, R = 2, B = 8, L = 5, Centre = 66
Wheel 3: T = 6, R = 4, B = 9, L = 2, Centre = ?

Visual Schematic / Puzzle Diagram:

```
     Wheel 1             Wheel 2             Wheel 3
       [ 4 ]               [ 7 ]               [ 6 ]
         |                   |                   |
[ 3 ]-( 39 )-[ 5 ]   [ 5 ]-( 66 )-[ 2 ]   [ 2 ]-(  ?  )-[ 4 ]
         |                   |                   |
       [ 6 ]               [ 8 ]               [ 9 ]
```

Visual Schematic / Puzzle Diagram:

```
     Wheel 1             Wheel 2             Wheel 3
       [ 4 ]               [ 7 ]               [ 6 ]
         |                   |                   |
[ 3 ]-( 39 )-[ 5 ]   [ 5 ]-( 66 )-[ 2 ]   [ 2 ]-(  ?  )-[ 4 ]
         |                   |                   |
       [ 6 ]               [ 8 ]               [ 9 ]
```

Visual Schematic / Puzzle Diagram:

```
     Wheel 1             Wheel 2             Wheel 3
       [ 4 ]               [ 7 ]               [ 6 ]
         |                   |                   |
[ 3 ]-( 39 )-[ 5 ]   [ 5 ]-( 66 )-[ 2 ]   [ 2 ]-(  ?  )-[ 4 ]
         |                   |                   |
       [ 6 ]               [ 8 ]               [ 9 ]
```

Visual Schematic / Puzzle Diagram:

```
     Wheel 1             Wheel 2             Wheel 3
       [ 4 ]               [ 7 ]               [ 6 ]
         |                   |                   |
[ 3 ]-( 39 )-[ 5 ]   [ 5 ]-( 66 )-[ 2 ]   [ 2 ]-(  ?  )-[ 4 ]
         |                   |                   |
       [ 6 ]               [ 8 ]               [ 9 ]
```

- **A** 58
- **B** 62

C 66

D 54

✓ **Verified Answer:** (B)** — `62` **Detailed Solution & Formula Deduction:** 62. Rule: Centre = (T+R)×(B+L) − (T×L + R×B). Wheel 1: (4+5)(6+3)−(4×3+5×6) = 81−42 = 39 ✓. Wheel 2: (7+2)(8+5)−(7×5+2×8) = 117−51 = 66 ✓. Wheel 3: (6+4)(9+2)−(6×2+4×9) = 110−48 = 62.

Problem 15 • Infosys SE Question 15

CORE SE

ID: Q15 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q53)
Section: Section 5: Numeric Puzzle Solving | **Page:** 7
Visual Reference Plate: paper_04_page_07.png

Puzzle Statement:
Matrix logic. Find the missing number.
2
3
4
6
8
10
12
15
?

Visual Schematic / Puzzle Diagram:

```
┌────┬────┬────┐
│  2 │  3 │  4 │
├────┼────┼────┤
│  6 │  8 │ 10 │
├────┼────┼────┤
│ 12 │ 15 │  ? │
└────┴────┴────┘
```

Visual Schematic / Puzzle Diagram:

```
┌────┬────┬────┐
│  2 │  3 │  4 │
├────┼────┼────┤
│  6 │  8 │ 10 │
├────┼────┼────┤
│ 12 │ 15 │  ? │
└────┴────┴────┘
```

Visual Schematic / Puzzle Diagram:

```
┌────┬────┬────┐
│  2 │  3 │  4 │
├────┼────┼────┤
│  6 │  8 │ 10 │
├────┼────┼────┤
│ 12 │ 15 │  ? │
└────┴────┴────┘
```

Visual Schematic / Puzzle Diagram:

```
┌────┬────┬────┐
│  2 │  3 │  4 │
├────┼────┼────┤
│  6 │  8 │ 10 │
├────┼────┼────┤
```

```
| 12 | 15 |  ?  |
└────┴────┴─────┘
```

A 16

B 18

C 20

D 21

✓ **Verified Answer:** (B)** — `18` **Detailed Solution & Formula Deduction:** 18. Rule: cell(row r, col c) = r×c + r^2 (rows and columns numbered from 1). Row 3: (3×1+9, 3×2+9, 3×3+9) = (12, 15, 18).

Problem 16 • Infosys SE Question 16

CORE SE

ID: Q16 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q54)
Section: Section 5: Numeric Puzzle Solving | **Page:** 7
Visual Reference Plate: paper_04_page_07.png

Puzzle Statement:
Pattern. Find the next term: 1, 5, 14, 30, 55, ?

Visual Schematic / Puzzle Diagram:

```
Terms:        1        5       14       30       55       [ 91 ]
              |        |        |        |        |          |
Sum:         1²      1²+2²   1²+2²+3²  1²+..+4²  1²+..+5²   1²+..+6²
Addends:     +1       +4       +9      +16      +25        +36
```

Visual Schematic / Puzzle Diagram:

```
Terms:        1        5       14       30       55       [ 91 ]
              |        |        |        |        |          |
Sum:         1²      1²+2²   1²+2²+3²  1²+..+4²  1²+..+5²   1²+..+6²
Addends:     +1       +4       +9      +16      +25        +36
```

Visual Schematic / Puzzle Diagram:

```
Terms:        1        5       14       30       55       [ 91 ]
              |        |        |        |        |          |
Sum:         1²      1²+2²   1²+2²+3²  1²+..+4²  1²+..+5²   1²+..+6²
Addends:     +1       +4       +9      +16      +25        +36
```

Visual Schematic / Puzzle Diagram:

```
Terms:        1        5       14       30       55       [ 91 ]
              |        |        |        |        |          |
Sum:         1²      1²+2²   1²+2²+3²  1²+..+4²  1²+..+5²   1²+..+6²
Addends:     +1       +4       +9      +16      +25        +36
```

A 91

B 85

C 96

D 90

✓ **Verified Answer:** (A)** — `91` **Detailed Solution & Formula Deduction:** 91. Each term is the sum of squares $1^2+2^2+...+n^2$: 1, 1+4=5, +9=14, +16=30, +25=55, next +36 = 91. Section 6: Grammar

Problem 17 • Infosys SE Question 17

CORE SE

ID: Q17 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q51)
Section: Section 5: Numeric Puzzle Solving | **Page:** 7
Visual Reference Plate: paper_05_page_07.png

Puzzle Statement:
Matchstick. Five unit squares are arranged in a plus/cross pattern (one centre square with one square attached to each of its four sides), using 16 matchsticks in total. What is the minimum number of sticks that must be removed so that no complete square remains?

Visual Schematic / Puzzle Diagram:

```
    +---+
    | 1 |
+---+---+---+
| 2 | 3 | 4 |
+---+---+---+
    | 5 |
    +---+
Plus / cross pattern of 5 squares made of 16 matchsticks.
```

Visual Schematic / Puzzle Diagram:

```
    +---+
    | 1 |
+---+---+---+
| 2 | 3 | 4 |
+---+---+---+
    | 5 |
    +---+
Plus / cross pattern of 5 squares made of 16 matchsticks.
```

Visual Schematic / Puzzle Diagram:

```
    +---+
    | 1 |
+---+---+---+
| 2 | 3 | 4 |
+---+---+---+
    | 5 |
    +---+
Plus / cross pattern of 5 squares made of 16 matchsticks.
```

Visual Schematic / Puzzle Diagram:

```
    +---+
    | 1 |
+---+---+---+
| 2 | 3 | 4 |
+---+---+---+
    | 5 |
    +---+
Plus / cross pattern of 5 squares made of 16 matchsticks.
```

A 3

B 4

C 5

D 6

✓ **Verified Answer:** (B)** — `4` **Detailed Solution & Formula Deduction:** 4. Each of the 4 outer squares shares exactly one stick with the centre square and no sticks with each other. Removing one shared stick breaks both the centre and that one arm, but each arm needs its own dedicated removal (no single stick touches two different arms), so at least 4 removals are required — and 4 suffice (remove all 4 centre-arm shared sticks). Verified by exhaustive computer search over all 16 sticks and 5 squares.

Problem 18 • Infosys SE Question 18

CORE SE

ID: Q18 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q52)
Section: Section 5: Numeric Puzzle Solving | **Page:** 7
Visual Reference Plate: paper_05_page_07.png

Puzzle Statement:
Circular wheel. Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.
Wheel 1: T = 8, R = 2, B = 6, L = 3, Centre = 41
Wheel 2: T = 10, R = 4, B = 5, L = 2, Centre = 65
Wheel 3: T = 12, R = 3, B = 9, L = 5, Centre = ?

Visual Schematic / Puzzle Diagram:

```
    Wheel 1            Wheel 2            Wheel 3
      [ 8 ]              [ 10 ]             [ 12 ]
        |                  |                  |
[ 3 ]-( 41 )-[ 2 ]  [ 2 ]-( 65 )-[ 4 ]  [ 5 ]-(  ?  )-[ 3 ]
        |                  |                  |
      [ 6 ]              [ 5 ]              [ 9 ]
```

Visual Schematic / Puzzle Diagram:

```
    Wheel 1            Wheel 2            Wheel 3
      [ 8 ]              [ 10 ]             [ 12 ]
        |                  |                  |
[ 3 ]-( 41 )-[ 2 ]  [ 2 ]-( 65 )-[ 4 ]  [ 5 ]-(  ?  )-[ 3 ]
        |                  |                  |
      [ 6 ]              [ 5 ]              [ 9 ]
```

Visual Schematic / Puzzle Diagram:

```
    Wheel 1            Wheel 2            Wheel 3
      [ 8 ]              [ 10 ]             [ 12 ]
        |                  |                  |
[ 3 ]-( 41 )-[ 2 ]  [ 2 ]-( 65 )-[ 4 ]  [ 5 ]-(  ?  )-[ 3 ]
        |                  |                  |
      [ 6 ]              [ 5 ]              [ 9 ]
```

Visual Schematic / Puzzle Diagram:

```
    Wheel 1            Wheel 2            Wheel 3
      [ 8 ]              [ 10 ]             [ 12 ]
        |                  |                  |
[ 3 ]-( 41 )-[ 2 ]  [ 2 ]-( 65 )-[ 4 ]  [ 5 ]-(  ?  )-[ 3 ]
        |                  |                  |
      [ 6 ]              [ 5 ]              [ 9 ]
```

A 85

B 81

C 90

D 78

✓ **Verified Answer:** (A)** — `85` **Detailed Solution & Formula Deduction:** 85. Rule: Centre = $(T-L)^2 + (B-R)^2$. Wheel 1: $(8-3)^2+(6-2)^2 = 25+16 = 41$ ✓. Wheel 2: $(10-2)^2+(5-4)^2 = 64+1 = 65$ ✓. Wheel 3: $(12-5)^2+(9-3)^2 = 49+36 = 85$.

Problem 19 • Infosys SE Question 19

CORE SE

ID: Q19 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q53)
Section: Section 5: Numeric Puzzle Solving | **Page:** 7
Visual Reference Plate: paper_05_page_07.png

Puzzle Statement:
Matrix logic. Find the missing number.
3
7
13
7
12
19
13
19
?

Visual Schematic / Puzzle Diagram:



Visual Schematic / Puzzle Diagram:



Visual Schematic / Puzzle Diagram:



Visual Schematic / Puzzle Diagram:



```
| 13 | 19 |  ?  |
└────┴────┴─────┘
```

A 25

B 27

C 29

D 24

✓ **Verified Answer:** (B)** — `27` **Detailed Solution & Formula Deduction:** 27. Rule: cell(r,c) = $r^2+c^2+r\times c$ (symmetric in r and c, so the matrix itself is symmetric about its diagonal — note how cell(1,2)=cell(2,1)=7). Row 3, column 3: $3^2+3^2+3\times3 = 9+9+9 = 27$.